



National review of hurricane evacuation plans and policies: a comparison and contrast of state practices

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Received 27 July 2001; received in revised form 22 April 2002; accepted 26 April 2002

Abstract

The events of recent hurricane seasons have made evacuation a leading emergency management issue. In 1998 and 1999, Hurricanes Georges and Floyd precipitated the two largest evacuations in the history of the United States and perhaps, its two largest traffic jams. In response to the problems experienced during these events, many state departments' of transportation have begun to take a more active role in the planning, management, and operation of hurricane evacuations. This is somewhat of a departure from prior practice when emergency management officials directed these tasks almost exclusively. Since the involvement of transportation professionals in the field of evacuation has been a fairly recent development, many of the newest practices and policies have only been used once, if ever. They also vary widely from state-to-state. To determine what the latest policies and strategies are and how they differed from one location to another, a national review of evacuation plans and practices was recently undertaken. The study was carried out from a transportation perspective and included both a review of the traditional transportation literature and a survey of department of transportation and emergency management officials in coastal states threatened by hurricanes. This paper highlights the findings of the survey portion of the study. It focuses mainly on current state practices, including the use of reverse flow operations and intelligent transportation systems. It also summarizes current evacuation management policies, methods of information exchange, and decision-making criteria. This paper presents the general similarities and differences in practices and gives particular attention to unique, innovative, and potentially useful practices used in individual states.

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Keywords: Evacuation; Emergency and disaster planning; Contraflow; Traffic engineering

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1. Introduction

Recent trends have increased the vulnerability of the United States (US) to hurricanes. The combination of growing population and development in coastal zones, rising ocean levels, coastal erosion, and changing climatic trends have increased the potential for loss of life and property in coastal regions of the country. A recent example of such an event was Hurricane Andrew. This massive storm struck both Florida and Louisiana in 1992, causing an estimated \$25 billion in damage and economic loss (USDOL, 1998). While more stringent building codes have been enacted to reduce damage from winds and flooding, not all coastal populations can be protected in their homes or shelters. To counter this threat, states in the Atlantic and Gulf coast regions have plans to evacuate people from vulnerable areas in advance of threatening storms.

Historically, hurricane evacuation planning has been the responsibility of emergency management (EM) and law enforcement agencies. While some state transportation agencies have contributed to the evacuation planning and management process, their activities could usually be characterized as peripheral support. In the two most recent large-scale evacuations, Hurricanes Georges in 1998 and Floyd in 1999, colossal traffic jams occurred revealing the fact that emergency response agencies were not as prepared for such scenarios as had been previously assumed. In the aftermath of these events, emergency management and state transportation officials recognized the need for an increased level of involvement from the transportation professional community.

Some of the needs to emerge from the Floyd and George experiences were ways to increase evacuation route capacity; the need to limit evacuation travel demand; the development of systems for better, faster, more reliable exchange of traffic flow and traveler information; and better planning for and coordination of regional and cross-state evacuations. To their credit, state departments of transportation (DOT) in every state threatened by hurricanes are now actively working at varying levels to better manage evacuations. In recent meetings and conferences organized for purpose of discussing plans and practices for evacuation, however, it was obvious that many DOT officials were not aware of the current state of evacuation practice nor the way in which new technologies and methods could be used to better address evacuation-related problems. This lack of awareness was not surprising given that most of them had never been involved in the evacuation planning and management process and there was little information available to them in the transportation literature.

To overcome this lack of information and the need for better information exchange, a national review of evacuation plans and practices in states threatened by hurricanes was undertaken. The study was carried out from a transportation perspective and included a review of literature from traditional transportation-oriented sources and a direct survey of DOT and EM officials. The information presented here focuses on the survey portion of the study and includes a general discussion of current practice; the use of reverse flow operations, intelligent transportation systems (ITS), and public transportation during evacuations; methods of information exchange; and EM policies and decision-making criteria. This paper also uses the survey responses to compare and contrast the general similarities and differences in practices, with particular attention given to unique, innovative, and potentially useful practices of individual states.

2. Survey background

Over the past five decades a wealth of information has been compiled from analyses of evacuations. Assessments of evacuation behavior have been carried out by the Federal Emergency Management Agency (FEMA) after most major hurricanes for several decades. More specific evacuation assessments, including the analysis of information related to transportation, have been prepared since the 1980's. While some information relating to development of evacuation planning and modeling can be found in traditional sources of transportation-oriented information, the vast majority of it is available in sources associated with the behavioral, geographic, sociologic sciences (for a comprehensive review of this literature, readers are encouraged to refer to the full project report (Urbina, 2002)). Thus, one of the difficulties in the recent effort to involve transportation professionals in the planning and management of evacuation was their lack of experience in the field and their unfamiliarity with sources of information on the subject. An additional complicating factor was that the development of the most innovative and potentially useful policies and practices for evacuation have occurred since the 1998 and 1999 storm seasons. As a result, not many of them are widely known and documented in the literature. To access the latest information, a direct survey of DOT and EM agencies was undertaken.

Surveys were sent to every at-risk coastal state. These included the 18 states on the Atlantic and Gulf coasts. A survey response form was also sent to emergency management officials in Hawaii. A total of 43 surveys were mailed, faxed, or emailed to EM, DOT, and law enforcement officials in each of the 19 states. At least one response was received from every state, for a total of 31 responses. Of these responses, 18 were from EM officials, 10 were from state DOT officials, and three were from law enforcement officials. Additional information was also gathered from officials at the Federal Highway Administration (FHWA), FEMA, private engineering firms, and academic researchers working in the field of evacuation. The following sections of this paper highlight the major findings of this survey. For a full discussion of the detailed aspects of the survey, including a list of the survey questions and the complete details of the survey responses, readers are encouraged to review the full project report (Urbina, 2002).

3. Evacuation plans and policies

The federal government, through FEMA, requires all states to have a comprehensive emergency operations plan. These plans guide emergency operations for all types of hazards, from natural to manmade and technological. And while the evacuation issues faced by coastal states are similar, the specifics of their plans are different since their geographic and transportation system characteristics vary widely.

Most states take a two-tiered approach to emergency planning and response. For the most part, evacuation planning, response, and recovery activities are developed at the local (e.g. county or city) level. State-level EM agencies typically served to coordinate local EM agencies, law enforcement, and transportation agencies. An example of the merit of this approach was illustrated by practices in the State of Texas where possible threats differed widely within the various regions of the state. The Texas State EM Plan included a general evacuation plan, however, specific

hurricane evacuation planning was left to local coastal jurisdictions. Plans for local inland jurisdictions in Texas were concerned more with sheltering and mass care issues rather than the movement of evacuees.

In Florida, where the entire state is vulnerable to hurricanes, the State EM agency took a greater managerial role in developing evacuation plans. Although despite the additional effort, events of recent evacuations showed that plans were not suitable to accommodate regional and multi-state evacuations. State plans also differed in the level of detail. The North Carolina Department of Transportation (NCDOT) is currently developing an Emergency Operations and Procedures Manual for the Division of Highways (DOH). When final, the plan will include information on various emergency response and recovery issues covering details down to debris removal, personnel and equipment needs, etc.

3.1. Authority and command structure

The emergency response command structure also differed in every state. By law, Governors in most states had the ultimate authority to order evacuations. However, some Governors delegated this authority to local-level officials, such as a mayor, city council, county sheriff, county judge, or county president. This was primarily because local level managers have a better knowledge of local characteristics and are better informed on current local conditions. It was also because evacuations tend to be unpopular and politically sensitive issues. Elected officials are not eager to order an evacuation because they could turn out to be unneeded, insufficiently large, or worse, insufficiently small.

Table 1 shows the evacuation ordering authority in the surveyed states. Governors maintained their authority in 15 of 16 responding states. In New York and Texas, authority was been delegated exclusively to local authorities. The stated reason for this was that only relatively small areas of these states are vulnerable to hurricanes. In South Carolina, the opposite was true, the Governor had sole authority to order a hurricane evacuation and would not delegated it to anyone else.

3.2. Evacuation implementation time

A critical issue in avoiding casualties from hurricanes is the time at which state or local authorities must order an evacuation. The earlier the evacuation order is issued, the more time residents and tourists will have to evacuate. Unfortunately, the earlier the evacuation order is issued the greater the possibility the hurricane could change course before landfall, rendering the evacuation unnecessary or leading evacuees to more dangerous locations. Evacuations that turn out to be unnecessary also lead to a “Cry Wolf” syndrome in which people are less likely to evacuate during future threats.

The primary criteria used to make decisions about how soon and how much area to evacuate were the storm forecasts issued by the National Hurricane Center (NHC). The NHC provides information on storm track, forward speed, and intensity to state and local authorities about every 6 h. Currently, the NHC forecasting system is able to predict strike locations within about 100 miles in the period 24 h before the expected storm landfall. Most of the agencies that

Table 1
Hurricane evacuation declaration authority

State	State agencies			Local agencies					
	Governor	State emergency management office	Department of public safety state police	Local emergency management office	Mayor	Highest local elected official	County or city law enforcement	County judge	County president
New Hampshire	×								
Massachusetts	×					×			
Rhode Island	×				×	×			
Connecticut	×		×		×		×		
New York						×			
New Jersey	×	×	×	×	×		×		
Delaware	×								
Maryland	×				×	×			
Virginia	×			×					
North Carolina	×								×
South Carolina	×								
Georgia	×				×				×
Florida	×								×
Mississippi	×				×				×
Louisiana	×								×
Texas					×			×	
Hawaii	×	×			×				

participated in the survey felt, however, that NHC predictions were neither timely nor accurate enough since even medium-sized coastal cities need 12 h to initiate and complete evacuations before arrival of tropical storm-force winds (39 miles/h).

The time required to evacuate is estimated from a combination of clearance times and the pre-landfall hazard time. Hazards times suggest how far in advance of the arrival of tropical storm-force winds a decision has to be made to allow a complete evacuation before hazardous conditions arrive. Clearance time is the time required to configure all traffic control elements on the evacuation routes, initiate the evacuation, and clear the routes of vehicles once deteriorating conditions warrant its end. Clearance times are estimated using data such as the population anticipated to evacuate, the number of lanes available for evacuation and impacts from other areas that will affect the evacuation. The amount of time required for clearance can be significantly lengthened by en route congestion and the setup time required for more complex control elements (such as those required for contraflow).

With all of the factors that need to be considered for an evacuation, it is not surprising that stated evacuation times also vary widely by location. These variables effect the minimum advance

Table 2

Evacuation advanced notification time (in hours)

State	Category of the storm				
	1	2	3	4	5
Massachusetts	9	9	12	12	12
Rhode Island	12–24	12–24	12–24	12–24	12–24
Maryland	20	20	20	20	20
Virginia	12	18	24	27	27
South Carolina	24	24	32	32	32
Georgia	24–36	24–36	24–36	24–36	24–36
Mississippi	12	24	24	48	48
Louisiana	24	48	72	72	72

times that states use to issue an evacuation order, as shown in Table 2. Most of these times were around 24 h. However, states with the need to evacuate large cities with limited outbound route capacities, like Louisiana, required a minimum of 72 h advanced notification time. Obviously, this is extremely difficult with the limitations of current weather forecasting.

3.3. Evacuation type

Once an evacuation is deemed warranted, the extent and type of evacuation must also be determined. The type and urgency is dependent on the characteristics of the storm and clearance times. Typically, evacuations are classified as one of three types: “voluntary”, “recommended”, and “mandatory”. In Louisiana, “voluntary” evacuations are often targeted toward people most vulnerable to hurricane surge and wind, including offshore workers and persons on coastal islands or in wetland areas. No special traffic control or transportation measures are usually taken during voluntary evacuations and people may remain if they choose. “Recommended” evacuations are issued when a storm has a high probability of causing a threat to people living in at-risk areas. Again, decisions of whether or not to leave are left to individuals and few special transportation arrangements are made. “Mandatory” evacuations are the most serious. During a “mandatory” evacuation, authorities put maximum emphasis on encouraging evacuation and limiting ingress to coastal areas. It is also the time when most evacuation transportation plans go into effect.

One of reasons that many states do not use “mandatory” evacuations is that they are difficult to enforce and many people resist being ordered to leave their homes and property by government officials. Although Mississippi had mandatory evacuations, its current laws do not allow State officials to enforce them. In the Florida Keys, Monroe County could legally declare a mandatory evacuation, however, practically speaking, they could not be enforced. New Jersey emergency management officials also acknowledged that if a person wanted to stay, the State would not physically remove them unless it was absolutely certain that they would be harmed.

There are also variations in the specific definitions and use of evacuation terminology between states. Virginia classified all evacuations as either “voluntary” or “recommended”, none were “mandatory”. In South Carolina, all evacuations were either “voluntary” or “mandatory”. In Texas there were no “mandatory” evacuations, though all evacuation were regarded to be “recommended”. In areas of Georgia susceptible to storm surge, evacuations could be “partial voluntary”, when only a portion of a county was asked to evacuate. However, Georgia also used

the terms “full voluntary”, “partial mandatory”, and “full mandatory” evacuation. In Jefferson Parish Louisiana, “voluntary” evacuations were referred to as “precautionary”.

The reason why the definition and terminology of evacuation declarations are important is because they impact peoples’ decision of whether or not to leave. Prior research has shown that people who said they heard mandatory evacuation orders are the most likely to evacuate; while recommended evacuation orders are met with less urgency (PBS&J, 2000a). The type of evacuation order and how it is communicated is also critical to avoid unnecessary evacuations. This phenomenon, known as “shadow evacuation”, occurs when people near threatened areas evacuate their homes even though they are not necessarily in danger. Authorities in Florida felt that one of the reasons for extreme number of evacuees during Hurricane Floyd was shadow evacuation. Thus, it is important for authorities to strike a balance between alerting those for whom evacuations are intended, while avoiding those for whom they are not.

3.4. Low mobility groups and use of public transportations for evacuations

Transportation infrastructure in the US has developed to serve vehicular traffic. While reliance on personal transportation works reasonably well under routine conditions, it can cause significant challenges for EM officials. Estimates of people without access to transportation in New Orleans, for example, ranged as high as 25–30% of the population. In addition to people without vehicles, potential evacuees also included the indigent, elderly, prisoners, the infirm, and tourists. The movement of people within these groups is an area which has been largely un-addressed by DOT officials planning for evacuations.

In practice, the responsibility for the movement of low mobility groups in facilities like prisons, hospitals, and schools is given to the chief administrator of the facility. Often, however, these administrators are neither familiar with, nor trained in, the procedures and regulations of emergency management or mass transportation. Recognizing this situation, some EMs have attempted to make special arrangements for these groups.

The most common mode of transportation for low mobility groups is busing. To transport people in busses EM agencies have in the past contracted with local transit authorities, school districts, and tour operators with varying levels of success. Additionally, many heavily populated cities do not have an adequate supply of busses to move all low-mobility evacuees. In New Orleans, for example, it was been estimated that about 250,000 residents (this does not include tourists or “special needs” populations) have no means of private transportation. Acquiring the number of busses required to transport this many people was not regarded to be practically feasible. Thus, Louisiana EM officials (like those in many other states) plan to use all available alternative means of transportation, including National Guard vehicles. Other forms of public transportation, such as rail or air services have been considered, though they were not widely regarded to be viable options.

4. New practices

The Georges and Floyd evacuations exposed many of the weaknesses of evacuation planning and management practices. Although some argue that the reason for the problems encountered in

these two evacuations were extreme and not likely to occur on a frequent basis, it is possible for evacuations of similar magnitudes to occur again. Regardless, the need for more effective transportation practices and technologies in evacuation is apparent and the expertise for them will have to come from the transportation community. The following sections summarize some of the ways that DOTs are working with EM agencies improve future evacuations through the application of contraflow operations and ITS systems.

4.1. Contraflow for evacuation

Contraflow, or reverse laning as it is also commonly known, involves the reversal of traffic flow in one or more of the inbound lanes (or shoulders) for use in the outbound direction with the goal of increasing capacity. In 1998 Florida and Georgia DOTs had plans in place to reverse the flow on their interstate freeways to expedite evacuation. Today, eleven of the 18 hurricane threatened states plan to use some type of contraflow evacuation strategy. For a more detailed discussion of the different types of contraflow evacuation operations and their associated benefits, costs, and inherent difficulties, readers are urged to refer to prior published reports (FEMA, 2000; PBS&J, 2000b; Wolshon, 2001).

While contraflow is widely viewed as the best way to increase outbound flow during evacuations, it is not universally considered to be the cure to evacuation problems. In fact, the true costs and benefits of contraflow in terms of its capacity improvements, safety, and manpower requirements remain largely unknown. Other concerns with contraflow evacuation include rerouting traffic and sheltering the enormous numbers of evacuees at the endpoint of the reversed segment. Contraflow was implemented for the first time in Georgia during Hurricane Floyd in 1999 with mixed, though overall positive, results. Contraflow was also improvised in South Carolina during Floyd after a strong public outcry came from evacuees trapped in congestion on I-26 from Charleston to Columbia.

Four different variants of contraflow are currently planned for use. They include: all lanes reversed; one lane reversed and one lane with normal inbound flow for emergency/service vehicle entry; one lane reversed with one lane normal flow for inbound traffic entry; and one lane reversed with the use of the shoulder of the outbound lanes. The various operational configurations and the states planning to use them are summarized in Table 3.

Because it offers the largest increase in capacity, the most common contraflow strategy is to reverse both inbound lanes to the outbound direction. A prior study estimated that a full reversal would provide a near 70% increase in capacity over conventional two outbound lane configurations (FEMA, 2000). A full reversal is also considered to be the least dangerous of the contraflow options because there is no inbound traffic to cause head-on accidents. However, at this time there are no research studies to back either of these assumptions.

Though not as widely planned, three other single-lane contraflow strategies are also proposed. Single inbound lane reversals are thought to increase outbound road capacity by about 30% (FEMA, 2000). Its main advantage, however, is its ability to maintain an inbound lane for re-entry of evacuees, law enforcement personnel, and emergency service vehicles. This is critical for clearing incidents and permitting access for returning evacuees. One of its major drawbacks, however, is that it could also raise the potential for head-on accidents.

Table 3
Evacuation contraflow use strategies

Strategy	State									
	New Jersey	Maryland	Virginia	North Carolina	South Carolina	Georgia	Florida	Alabama	Louisiana	Texas
Two lanes reversed		•	•	•	•	•	•	•	•	•
One lane reversed, one lane normal for emergency/service vehicle access	•								•	
One lane reversed, one lane normal for inbound traffic entry	•				•					
One lane reversed and use of outbound shoulder									•	

Notes: EM and DOT officials in the State of Delaware plan to use contraflow, although the specific configuration type has not yet been determined. Officials from the State of Mississippi are in discussions with their counterparts from Louisiana to extend to northbound I-59 contraflow out of New Orleans into their state.

Another variant of the one-lane strategy is to use of the right shoulder of the outbound lanes. This type of operation has been estimated to increase capacity by only about 8% (FEMA, 2000). Although the capacity increase from this configuration also depends on the width and condition of the shoulder, since flow rates are decreased and drivers tend to reduce speeds when they are laterally constrained. Two additional concerns associated with the use of shoulders are pavement suitability and bridge widths. Typically, shoulders are designed with a thinner pavement cross-section and greater cross-slope. Thus, there are concerns that they will not be able to withstand prolonged traffic loading and will provide an inadequate riding surface. Cross-section width can be a problem on bridges. Many freeway bridges, particularly older ones, have been constructed with narrow shoulders. If shoulders were used as an additional outbound lane, these locations would create bottlenecks with the potential to create additional congestion as vehicles merge back into the through lanes.

4.1.1. *Contraflow design*

Currently, there are no recognized standards or guidelines for the design, operation, and location of contraflow segments. Most contraflow designs have been developed from the application of standard design practice and past evacuation experiences. From the results of the survey it was apparent that while some of their geometric design and traffic control elements were similar, there were also many obvious differences.

The planned contraflow segments reviewed in this study typically began with a cross-over road constructed over a median or between interchange ramps to redirect a portion of the outbound traffic stream into the inbound lanes. Specific designs, however, varied by location. The most common method to affect the traffic split was through the use of a median cross-over. The specific location and design of these cross-over points was a function of roadway geometry, the approximate beginning of prior evacuation congestion, and the proximity of the route to other



Fig. 1. I-10 Median cross-over Kenner, Louisiana.

evacuation routes. A typical at-grade cross-over configuration is shown in Fig. 1. This location, in Kenner Louisiana, will divert traffic from the two left lanes of westbound I-10 into the eastbound lanes as evacuees depart New Orleans toward Baton Rouge. A water-filled segmented barrier has also been placed in the cross-over to prohibit vehicles from crossing the median during non-evacuation periods.

Cross-over designs at the interchange of two freeways can be more complex. An example of an interchange cross-over is shown in Fig. 2. At this location, a connecting roadway has been constructed between the loop ramp of I-526 to I-26 in Charleston South Carolina to divert traffic into the contraflow lanes of I-26. An additional connecting road is used to divert evacuation traffic from eastbound I-526 into the contraflow lanes of I-26 heading toward Columbia.

The contraflow section termini designs reviewed in the study also varied by location. One of the controlling criteria for the location of a termination point was the prevention of merging congestion. This was planned in several different ways. The method most commonly seen, particularly for shorter segments, was to permanently split the traffic flows. In this type of design one of the streams of traffic is diverted to a separate roadway, while the other continues travel on the original route. In New Orleans, traffic traveling in the normal outbound lanes of I-10 will all be diverted northbound to I-55 some 25 miles outside of the City. After the I-55 diversion, the traffic in the contraflow lanes of I-10 will cross the median back into the normal outbound lanes (Louisiana State Police, 2000). A similar design will also be used to terminate contraflow operations in South Carolina. At the end of the I-26 contraflow section in Columbia all evacuation

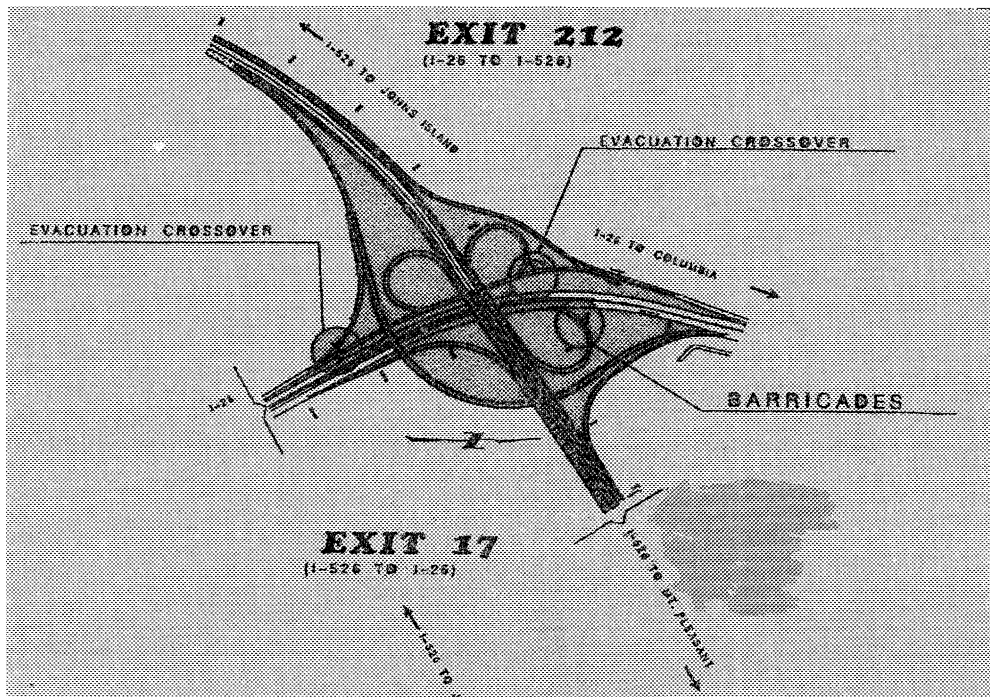


Fig. 2. I-26/I-526 interchange cross-overs Charleston, South Carolina.

traffic in the normal outbound (westbound) lanes will be diverted on to northbound I-77, while the I-26 contraflow traffic will be diverted back into the normal westbound lanes after the I-26/I-77 interchange.

Another common type of contraflow termini design was the “attrition-merge”. These designs were favored in states such as Georgia and Texas that have planned long contraflow segments. In this design, traffic in the normal and reverse flow lanes is reduced by allowing vehicles to exit to secondary routes at points along the contraflow segment. Through a process of exit attrition, it is assumed that traffic would be reduced to a level at the end of the segment that would allow a merging of the traffic streams without causing bottleneck congestion (Ford et al., 2000).

The lengths of contraflow segments were a function of the evacuation area geography and the road infrastructure. The segments ranged from as short as 3.5 miles to as long as 180 miles. Short sections have been planned to gain maximum capacity on routes that are between or connect other traffic arteries. Longer segments were planned to evacuate coastal cities to inland cities such as those between Corpus Christi and San Antonio, Charleston and Columbia, and Mobile and Montgomery, or for connecting to distant major evacuation highways. In Table 4 summarizes the planned routes and lengths for evacuation contraflow routes.

In Table 4 the extent to which some states plan to maximize the use of contraflow can also be seen. Florida had the most extensive plan for contraflow operations with seven freeway segments planned for use during evacuations. New Jersey had plans for five contraflow routes. Although in contrast to Florida, they were considerably shorter. Both of these states were also analyzing the

Table 4

Planned contraflow evacuation routes

State	Route(s)	Approximate distance (miles)	Origin location	Termination location
New Jersey	47/347	19	Dennis Twp.	Maurice River Twp.
	Atlantic city expressway	44	Atlantic City	Washington Twp.
	72/70	29.5	Ship bottom boro	Southampton
	35	3.5	Mantoloking boro	Pt. Pleasant Beach
	138/I-195	26	Wall Twp.	Upper Freehold
Maryland	MD-90	11	Ocean City, MD	US 50
Virginia	I-64	80	Hampton roads bridge	Richmond
North Carolina	I-40	90	Wilmington	Benson (I-95)
South Carolina	I-26	95	Charleston	Columbia
Georgia	I-16	120	Savannah	Dublin
Florida	I-10 Westbound	180	Jacksonville	Tallahassee
	I-10 Eastbound	180	Pensacola	Tallahassee
	SR 528 (Beeline)	20	SR 520	SR 417
	I-4 Eastbound	110	Tampa	Orange County
	I-75 Northbound	85	Charlotte County	I-275
	FL Turnpike	75	Ft. Pierce	Orlando
	I-75 (Alligator Alley)	100	Coast	Coast
Alabama	I-65	135	Mobile	Montgomery
Louisiana	I-10 Westbound	25	New Orleans	I-55
	I-10/I-59 (east/north)	115		Hattiesburg, MS
Texas	I-37	90	Corpus Christi	San Antonio

Note: Delaware and Virginia contraflow plans are still under development by their DOT.

feasibility of adding more contraflow routes, including I-75 from Wildwood to I-275 in Florida and the Garden State Parkway, New Jersey Turnpike, I-78 and I-80 in New Jersey (Trammell, 2001; Augustiniak, 2001).

An additional area of difference that was found during the review of contraflow designs was the extent to which certain states incorporated more detailed and differing aspects of the process. These variations appeared to be related to the agency that prepared the plan. In Alabama and North Carolina, where the plans were developed by the DOT, contraflow design plans included a considerable level of detail in the geometric design and traffic control aspects of the cross-over location. The North Carolina plans, for example, emphasized the placement and location of traffic control devices in the vicinity of the cross-over and the use of highway advisory radio (HAR) and dynamic message signs (DMS) messages to guide evacuees. The Louisiana plan, developed primarily by the State Police, paid considerably more attention to the enforcement requirements in the contraflow area. These plans detailed specific numbers and locations of police vehicles at the beginning and termination points of the segment and included the number of troopers and vehicles needed at each exit ramp.

4.1.2. Managing contraflow operations

In addition to the variation in designs, states also had significantly differing philosophies on how they planned to utilize and manage contraflow operations on their highways. South Carolina,

Table 5
Authority to start and end contraflow operations

State	Start	End
New Jersey	Governor	Governor
Delaware	Governor	Governor
Maryland	Local Emergency Management with the State Police & Maryland DOT	Local Emergency Management with the State Police & Maryland DOT
Virginia	Governor	Governor
North Carolina	Governor	Governor
South Carolina	Governor	Department of Public Safety
Georgia	Governor in conjunction with Georgia DOT and GEMA	Georgia DOT
Florida	Governor	Highway Patrol
Alabama	Alabama DOT	Alabama DOT
Louisiana	Governor	Governor
Texas	Highway Patrol Captain in Corpus Christi	Highway Patrol Captain in San Antonio

for example, was the only state that had plans to use contraflow for post storm re-entry. Additional differences were also apparent in the way states viewed the somewhat conflicting needs of having a rapid and orderly evacuation using contraflow versus the need to maintain acceptable levels of safety on the road. Despite the advantages that contraflow operations could bring to an evacuation, its disadvantages also need to be considered. All highway agencies agreed that reverse flow operations would likely be inconvenient and confusing for drivers. They also expected them to be labor intensive to initiate, difficult to enforce, and potentially dangerous for drivers. Survey questions were posed to determine these differences as well as managerial aspects such as who will decide when to use contraflow; under what conditions it will be started and ended; how long it will last; and how they plan to address issues associated with safety, accessibility, convenience, enforcement, and cost.

In most states the authority to control the start and end of contraflow operations resided with the Governor, although again, this varied. Table 5 shows the agency or individual who had the authority to initiate and terminate contraflow operations in the states where they are planned.

As the table also shows, several states have appointed DOT or State Police representatives to be in charge of their contraflow operations. Consistent with their philosophy of local-level evacuation authority, Texas has given operational control to local law enforcement officials at the beginning and ending of the segment. The only state where DOT officials had the authority to initiate and terminate contraflow operations was Alabama.

The controlling criteria that states cited as the basis for initiating contraflow operation were the storm characteristics (size and intensity) and forecast movement along with the time of day relative to the end of daylight. In cases where the storm was not forecast to make imminent landfall or was of modest strength, most states stated they would resist the use of contraflow. The other criterion that states would consider was traffic volume. Because of the inherent difficulties of its use, the majority of states felt that reverse flow lanes should not be implemented until traffic volumes warranted their use. Officials in these states stated their intention to wait until volumes were at, or rapidly approaching, capacity levels before using contraflow.

These opinions were not, however, shared by all states. Officials in the States of South Carolina and Louisiana both stated that they planned to initiate contraflow operations as soon as the call for an evacuation was made. It was their opinion that attempts to initiate contraflow operations after the normal outbound lanes are at, or about to reach, capacity would result in the loss of valuable evacuation time.

To initiate contraflow, devices and barricades must be erected, inbound lanes must be cleared of vehicles over their length, traffic control, and law enforcement and DOT field personnel need to be positioned at their assigned locations. Most states anticipated this process to take from 4 to 12 h depending on the length of the segment, the number of interchanges involved, and the number ramps and merge points that may require control. In a few states this process could take considerably longer. Authorities in Florida estimated that 49 h would be needed to prepare for a contraflow operation. The time is so much longer than other states because Florida will need to activate National Guard forces to set up and patrol their segments (Collins, 2001).

Because of these difficulties, limited visibility, and the need for additional enforcement personnel, most states are also reluctant to implement contraflow after nightfall. Florida officials stated that they would neither initiate nor operate any lane reversals at night. Georgia officials would also resist beginning contraflow operations after nightfall. However, they also recognized that some situations may dictate the need and they want to maintain flexibility in their response. By contrast, New Jersey and Maryland officials planned to implement contraflow regardless of the time of day.

4.1.3. Ending contraflow

The two most commonly reported factors used to determine the need to shutdown an evacuation (contraflow or otherwise) were the arrival of tropical storm force winds and an a decrease in evacuation traffic volumes. Most states also agreed that contraflow routes would remain open as long as weather conditions were favorable. Virginia planned to shutdown contraflow operations about 2 h prior to the forecast arrival of tropical force winds. This additional lead-time would allow them to evacuate DOT and law enforcement personnel. The State of North Carolina, by contrast, planned for a shutdown approximately 3 h before the same conditions. In Florida, the contraflow termination shutdown would be based solely on the arrival of nightfall. Georgia planned to maintain the contraflow operations continuously, until the storm threat had past.

While termination criteria differed, all states agreed that the most important consideration in any evacuation was the elimination for the potential for evacuees to be stranded on the highway. Despite this, the Palm Beach County (Florida) Division of Emergency Management had also planned for these cases in the event they were to occur. “Refuges of Last Resort” have been designated within a mile of certain evacuation routes in the county. These locations were for extreme circumstances and may, not necessarily have food, water, utilities, or supervision. They were not even guaranteed to be safe in strong hurricanes. However, they would provide evacuees a better option than remaining on the road during the storm.

4.2. Intelligent transportation systems for evacuation

Access to timely and accurate traffic information during evacuations is critical to evacuation process. Information about traffic flow rates and speeds along with lane closures, weather

conditions, travel times, incidents, and the availability of alternative routes is critical for the ability to give travel guidance to evacuees. During the Georges and Floyd evacuations, access to and exchange of accurate and timely traffic information was often difficult. During these events EM officials were in effect “working blind”, with little knowledge of which evacuation routes were flowing and which were in gridlock. As a result, they were unable to redirect traffic from routes that were over capacity to nearby roads that were carrying little traffic.

One of the ways that DOT officials were responding to the need for information is through the application of ITS technologies. All of the surveyed states had plans to enhance existing and develop new ITS systems for use in hurricane evacuations. The most common area of ITS application was for real-time monitoring of travel conditions. Several states, including South Carolina, Florida, and Louisiana, were either using or planning to use remote traffic detection systems.

All DOTs had some type of statewide traffic data recording system. They are used for planning purposes to monitor and assess statewide traffic volume and speed characteristics. While the design of individual systems varied, most used a remote sensing system (i.e., pavement loops) and a basic traffic data recorder. Under routine operation, most DOTs downloaded data from the recorders via telephone connections on an approximate monthly basis. Recently, minor modifications to these same systems have allowed South Carolina and Florida DOT officials to retrieve this information on hourly or 15-min bases during evacuations. The modifications will also allow data to be assembled and displayed in tables and graphs to monitor the progression of the evacuation, track volume changes, and identify routes with excess capacity. While this may not provide true real-time information, it will give EM and DOT evacuation coordinators a much better idea of up-to-date travel conditions than they have had in the past.

Other traffic surveillance systems planned for use include cameras and other advanced sensors capable of monitoring speed and flows. One of the most favored of these systems was closed circuit television (CCTV). CCTV cameras have an advantage over loop detection in that they can provide direct visual confirmation of traffic (and weather) conditions at remote locations. They can also be used for detecting incidents and verifying their removal. One of the limitations of these systems is that they also typically require direct power and communication connections. This is often difficult in remote locations along evacuation routes.

ITS systems are also planned to disseminate travel information to evacuees. Two of the systems planned for use are HAR and DMS. In contrast to traffic counter and CCTV systems that bring data in, HAR and DMS systems get information out. To make the most effective use of these tools (HAR has a limited range of about 3–5 miles), states were planning to use them in advance of exits and interchanges where services and alternative routes were available. The type of information conveyed through HAR and DMS included shelter locations, alternative evacuation routes, congestion, incident information, and services such as gas station, rest area locations, lodging availability among others. To increase its communication capabilities the Delaware DOT was in the process of acquiring a commercial FM radio station for use as a statewide travel information station. The station would convey general travel information during non-evacuation periods and evacuation information in advance of hurricanes. The relatively small area of Delaware also made it one of the few states where this was a viable option.

Another form of evacuation communication that has gained favor with DOTs is the Internet. The Internet can also be combined with ITS systems to enhance its capability. Several DOT and

EM agencies maintained websites to keep people informed of evacuation routes, road conditions, and weather information. The Virginia Department of Transportation (VDOT) had one of the most comprehensive sites for this purpose. The VDOT website included maps of all state evacuations routes, real-time traffic volume conditions, road incidents, and construction information. The site also provided links to emergency management and weather sites, including FEMA, NHC, Virginia Department of Emergency Services, and the Red Cross. The Florida Division of Emergency Management website provided links to hotels in Florida, Alabama, and Georgia that allowed evacuees to make online hotel reservations. The Florida DOT's website provided access to its statewide network of real-time traffic volume and speed data recorders. While all of these internet systems were targeted for use prior to the evacuation, there was little doubt that future wireless-Internet technologies will also allow them to be used en route.

There were also limitations in the application of ITS to hurricane evacuation. One of the most significant was that many ITS systems were located in urban areas, while the majority of evacuation route mileage was in rural areas. To address this, several states were planning to use portable systems, particularly HAR and DMS. Researchers in Louisiana were also working on the development of a mobile traffic data recording system. When completed, these self-contained mobile data collectors will be used to provide real-time traffic flow information using wireless communication technologies. Another advantage to mobile systems is that they can be deployed to any location in response to varying hurricane scenarios. One significant limitation of ITS is its expense. Since evacuations are such rare events that cover such wide areas, it is difficult to justify their cost unless they can incorporate multipurpose functionality.

5. Work zones

A historically overlooked issue in evacuation planning and preparedness has been highway work zones. In 1998, during the evacuation for Hurricane Georges, the States of Alabama, Mississippi, and Louisiana all had construction zones on evacuation routes. In Louisiana, evacuation traffic on westbound I-10 out of New Orleans was limited to a single lane. Early recognition of this problem by the DOT allowed them that to request the contractor clear construction equipment and open both of the partially constructed lanes to outbound traffic. Fortunately the contractor acted quickly and delay was minimized. However, things could have been worse had the storm not changed to an easterly course. A year later this similar problems of evacuation route construction were also seen during Hurricane Floyd in North Carolina.

Since the need for maintenance and construction during the hurricane season is unavoidable, some DOTs have made attempts to avoid conflicts by adding special provisions in construction contracts to accommodate evacuation traffic through work zones. The most common way to do this has been to add clauses that require a contractor to cease all construction activities once an evacuation is declared, clear all equipment, and open of all lanes of traffic even those under construction.

These types of restrictions were not usual. Most states, particularly those where traffic congestion is routine, restricted construction that reduced capacity. For example, the Maryland and New York DOTs did not allow construction to restrict traffic on any state arterial route to less than the normal number of lanes during the peak summer travel period from June to September.

While these restrictions covered most of the hurricane season they did not apply to the less active, though still potentially dangerous, months of October and November. In Texas, only the Houston District of the TxDOT included contract language that requires contractors to maintain the same number of lanes for evacuation as were originally available. In other TxDOT districts, this provision was understood, but not necessarily written into contracts.

Other suggestions that were made to maintain capacity through work zones on evacuation routes included limiting the construction season, length, contract duration, and/or phase sequencing of projects on evacuations routes. These types of contraction provisions can, however, also increase the cost and duration of projects since may require a contractor to work in shorter segments or use non-standard construction practices. They had also not been evaluated to determine how they would work during an evacuation.

6. Conclusion

This study showed encouraging signs for the improvement of hurricane evacuations. It was apparent that transportation professionals have become more involved in the development of evacuation plans since 1998. This is a significant step and represents a needed change from prior practice. The involvement of transportation planners and engineers has also brought expertise and experience in dealing with transportation-related issues, including forecasting evacuation travel demand; evacuation traffic analysis and modeling; and the application of ITS technologies.

However, many needs still exist. The two most evident are the needs to control evacuation demand and increase the capacity of evacuation routes. From the survey and discussions at numerous conferences, it is apparent that no amount of road building or increased efficiency will permit the complete evacuation of major cities like Houston, Miami, and New Orleans in one or two days. Thus, it becomes necessary (using in-place sheltering and other means) to limit the amount of evacuees and use the existing highway network to its maximum efficiency. This includes the use of contraflow on limited access evacuation routes, the coordination of traffic control systems on arterial routes, and the use of mass transportation assets, among others.

There are also practical and operational needs. One of these is for a greater and more efficient exchange of data and information. This includes remote traffic speed and flow data for the monitoring, assessment and management of the evacuation and better communication of guidance information from emergency management officials to the evacuees. There is also a need to develop a standard set of evacuation guidelines. While it is recognized that evacuation plans need to allow room for flexibility, checklists and basic guidelines could benefit state and local level DOT and EM officials in the development of specific plans. They would also ensure a basic level of practice uniformity from location to location; an issue has gained importance as evacuations become regionalized. There is also a need to address specific transportation related evacuations issues, including those associated with the movement of low-mobility groups, accommodating evacuation traffic through work zones, and determining the costs and benefits of contraflow operations. Currently, there are no studies that have addressed these issues from an analytical standpoint.

One of many positive outcomes of the Floyd experience is that the FHWA has recognized many of these needs and has initiated a series of regional evacuation-related workshops and established

a Hurricane Evacuation Liaison Team that will work with FEMA to improve coordination and communications during hurricane events. A Transportation Research Board subcommittee on evacuation has also been formed to bring attention to evacuation needs and issues, support the development of future research initiatives, and disseminate new research findings to the practicing community. Over the long term it is hoped that research initiatives will be undertaken to address the variety of needs that currently exist. Some of the most critical are related to evacuation travel demand behavior; cost/benefit analyses for evacuations strategies, including the costs and benefits of contraflow, use of secondary routes, intermediate cross-overs for contraflow operations, and the installation of evacuation traffic signs and ITS devices; the application of web-based systems for the communication of real time traffic conditions, availability of shelters, and evacuee route guidance; effective methods for post evacuation re-entry; development of micro and macro-level traffic models to analyze evacuation bottlenecks to evaluate different storm scenarios, routing options, and contraflow strategies; and planning for the effective use of mass transit during evacuations for special needs populations and low mobility groups such as tourists and homeless.

Acknowledgements

The authors would like to acknowledge the technical assistance and information provided to them by Dr. Marc Levitan, Director of the LSU Hurricane Center and Dr. Chester Wilmot of the LSU Department of Civil and Environmental Engineering. The authors also express their gratitude to all of survey respondents for the information they provided and to the anonymous reviewers for their constructive comments given during the preparation of this paper.

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